



Data Analytics Project

- 1) Predicting Heat Dissipation in Mechanical Assemblies (Regression)
- 2) Predicting Machine Failure in a Manufacturing Plant



1

What we find and goals

Predict the heat lost during operation

We Build the suitable python code, algorithm using data set to predict and analysis the heat loss during the Operation

Contents in our data set

- **Material thermal**
- **Conductivity**
- **Surface area**
- **Contact pressure**
- **Heat source power**
- **Cooling method**
- **Time**
- **Operating temperature**
- **Assembly type**
- **Thickness**
- **Ambient temperature**

ALGORTHIM

The list of 6 Algothims are used to predict the heat loss



LinearRegression

Score: **0.51**



Ridge

Score: **0.27**



ElasticNet

Score: **0.51**



DecisionTreeRegressor

Score: **1.0**



MLPRegressor

Score: **-0.3**



Laso

Score: **0.2**

The Best Algrothim Used

Using this Best Algrothim to predict the heat loss during the Operation

DecisionTreeRegressor

Import code: **from** sklearn.tree **import** DecisionTreeRegressor

Score: **1.0**

What are the Benefits are

- **No need for physics equations – learns from data**
- **Fast predictions – quick after training.**
- **Handles non-linear patterns – good for complex systems**
- **Interpretable – tree structure is easy to understand.**
- **Feature importance – shows what affects heat loss most.**
- **Adaptable – improves with more data.**

2

What we find and goals

Predicting Machine Failure in a Manufacturing Plant

We Build the suitable python code, algorithm using data set to predict and analysis the Machine failure in the manufacturing plant.

Contents in
our data set

Feature Name	Description
Temperature	Internal machine temperature (°C or °F) – overheating can be a failure sign.
Pressure	Hydraulic or pneumatic pressure – abnormal values may indicate malfunction.
Vibration	Mechanical vibration levels – high values often signal component wear.
RPM	Rotations per minute – drop or spike may indicate motor problems.
Power consumption	Amount of electricity consumed – irregularity can imply mechanical resistance.
Lubricant level	Oil/grease level – too low = friction & overheating risk.
Humidity	Environmental humidity – affects electrical components.

ALGORTHIM

The list of 6 Algothims are used to predict the heat loss



LinearRegression

Ridge

ElasticNet

RandomForestRegressor

SVR

Laso

Score:

Score:

Score:

Score:

Score:

Score:

0.011

0.011

0.000654

0.838

-0.0004

0.00037

The Best **Algrothim** Used

Using this Best Algrothim to predict the Defect in the manufacturing plant

RandomForestRegressor

Import code: **from** sklearn.ensemble import **RandomForestRegressor**

Score: **0.838**

What are the Benefits are

✓ 1. Early Warning System

Detect risky patterns and warn about machines that are likely to fail soon.

✓ 2. Reduce Downtime

If the system predicts a potential failure, the machine can be serviced before breakdown, saving production time.

✓ 3. Cost-Efficient Maintenance

Avoid unnecessary routine maintenance and focus on machines that actually need attention, reducing labor and spare parts costs.

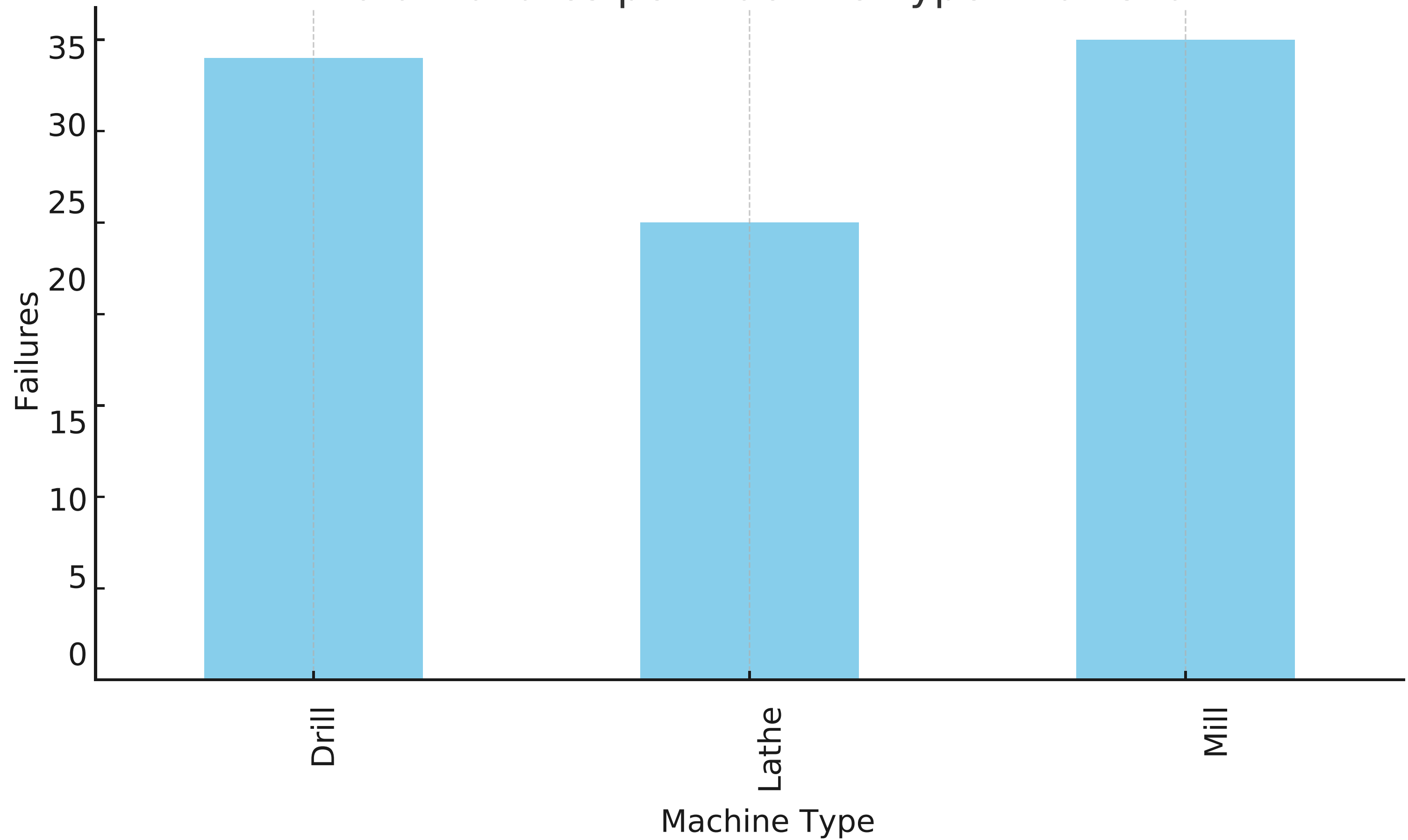
✓ 4. Improve Safety

A failing machine can be dangerous. Predicting failure avoids accidents and ensures worker safety.

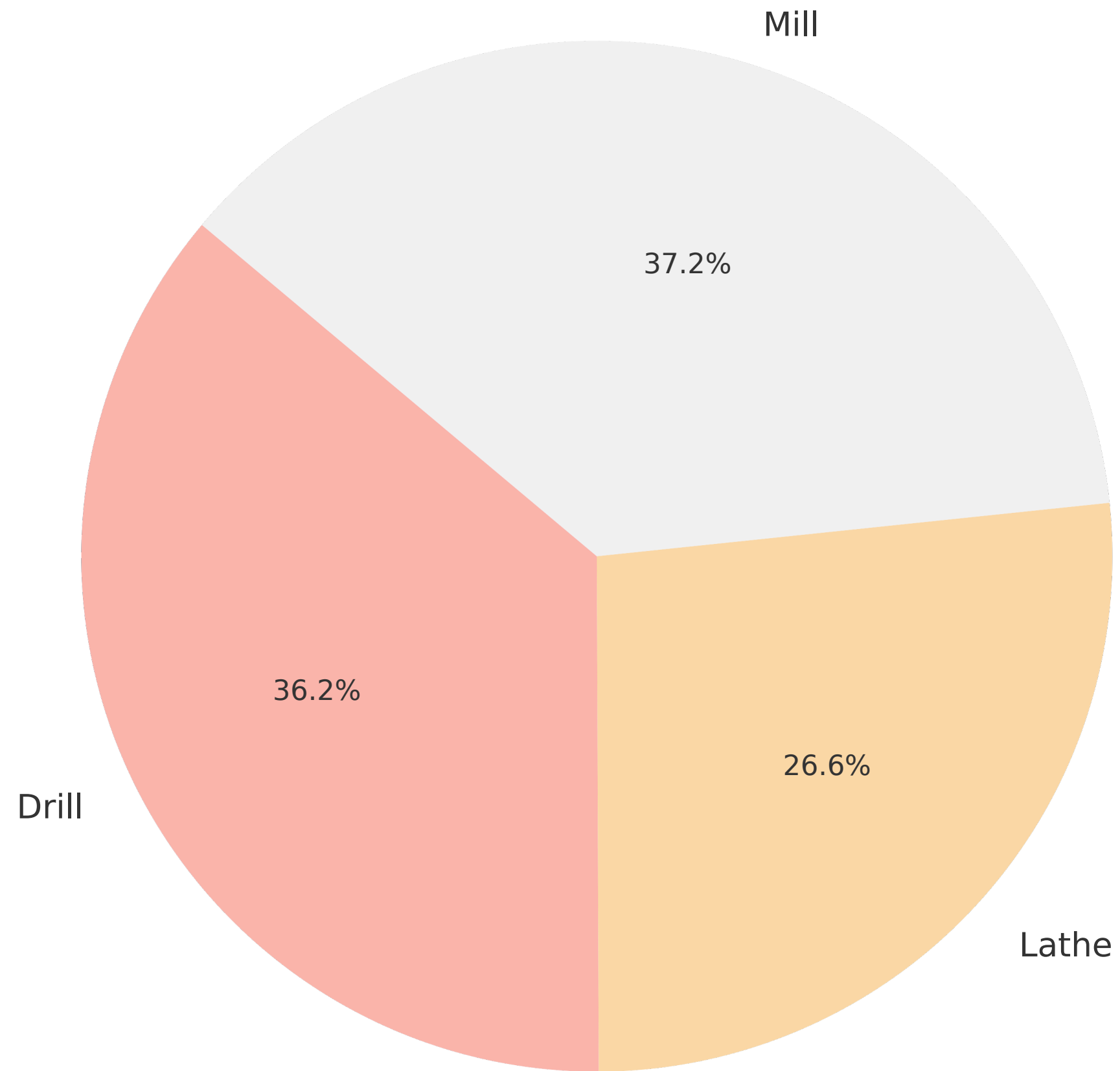
✓ 5. Data-Driven Decisions

Helps the maintenance and operations team to make decisions based on real-time insights, not just intuition.

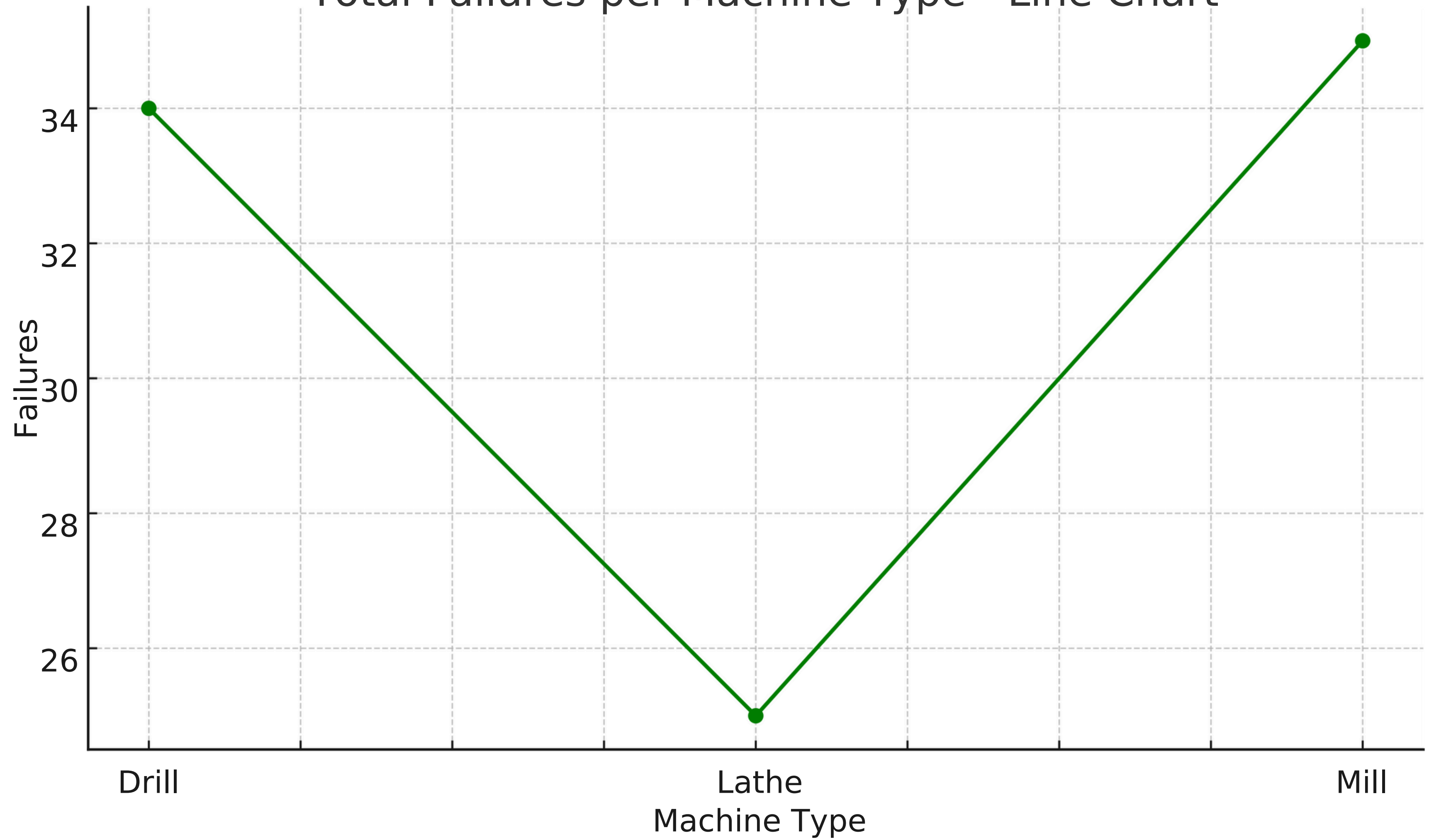
Total Failures per Machine Type - Bar Chart

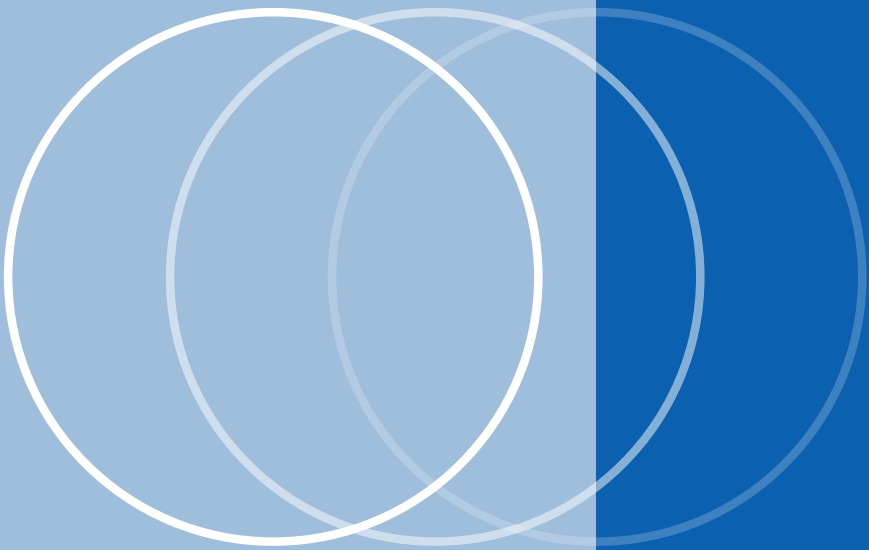


Failure Distribution by Machine Type - Pie Chart



Total Failures per Machine Type - Line Chart





Thankyou

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